Reaction-Driven Self-Oscillation in Thermoresponsive Gels: From Bifurcation Analysis to Geometry-Dependent Dynamics

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(Dated: April 14, 2026)

Thermoresponsive hydrogels with lower critical solution temperature (LCST) can exhibit self-sustained volume oscillations when coupled to an exothermic chemical reaction. We develop a hierarchy of models—from a lumped zero-dimensional (0D) system to spatially resolved one-dimensional (1D) slab and spherical geometries—to elucidate the oscillation mechanism and its dependence on geometry. The 0D model identifies Hopf bifurcation boundaries in the Damköhler number Da and LCST sensitivity S_χ parameter space, predicting two oscillatory regimes: a large-amplitude relaxation oscillator (Regime I) and a small-amplitude near-harmonic oscillator (Regime II). The 1D slab model reveals the spatial structure of these oscillations, including a dual-zone termination mechanism in which the outer shell is shut off by accessibility collapse while the inner core is limited by reactant depletion. Crucially, we show that geometry plays a decisive role: the slab geometry sustains robust oscillations, whereas the spherical geometry suppresses them due to enhanced radial solvent influx and, in the full-mechanics case, anisotropic stress locking. Finally, we demonstrate that reactant penetration depth can be enhanced more than 590× by optimizing the Da–D_0–Bi_c parameter combination, providing design guidelines for autonomous chemical-delivery applications.

I. INTRODUCTION

Stimuli-responsive gels that undergo spontaneous, self-sustained volume oscillations have attracted attention as minimal models of biological rhythms and as candidate components for soft actuators, chemical oscillators, and autonomous drug-delivery systems [1, 4]. The most studied synthetic realization couples a thermoresponsive network—typically based on poly(N-isopropylacrylamide) (PNIPAM) with a lower critical solution temperature (LCST) around 33 °C—to an oscillatory or exothermic chemical reaction such as the Belousov–Zhabotinsky (BZ) reaction [2, 3]. In these systems, chemical oxidants produced by the BZ reaction swell the gel, while reductants cause collapse; the gel's spatiotemporal dynamics feed back on the local oxidant concentration, giving rise to autonomous oscillation [4].

A complementary oscillation route, less well studied, relies on purely *thermal* feedback: an exothermic reaction heats the gel above the LCST, the attendant volume collapse expels solvent and reactant, the gel cools, swells again, and re-ingests fresh reactant—completing one cycle. This thermo-mechanical-chemical feedback loop does not require multi-step BZ chemistry; a single irreversible exothermic reaction suffices. Early theoretical work by Siegel and coworkers demonstrated this concept for spherical gel beads [5, 6], and subsequent studies identified parameter regimes of self-oscillation in slab and membrane geometries [7, 8]. However, several fundamental questions remain open:

1. What are the minimal conditions—expressed in terms of dimensionless groups—for the thermal oscillation to occur?
2. How does the spatial structure of the gel affect the oscillation mechanism, and what role does geometry (slab vs. sphere) play?
3. Can the oscillation be harnessed for practical applications such as periodic chemical delivery, and if so, what parameters maximize interior reactant penetration?

In this paper, we address these questions through a systematic hierarchy of models:

- An *analytical linear-stability analysis of the 1D slab* (Sec. III) that derives the dispersion relation, identifies the Hopf window, and separates long-wave oscillatory instability from finite-wavenumber spinodal growth.
- A *1D slab model* (Sec. IV) that resolves spatial gradients and reveals the oscillation mechanism, including two distinct regimes and a dual-zone termination mechanism.
- A *1D sphere model* (Sec. V) that demonstrates how radial geometry suppresses oscillation via enhanced solvent transport and anisotropic stress effects.
- A *penetration optimization study* (Sec. VI) that provides quantitative design guidelines for maximizing interior reactant delivery.

The model integrates large-deformation poroelastic mechanics (Flory–Rehner free energy), heat diffusion with exothermic reaction source, and reactant transport coupled to the local swelling state. The dimensionless formulation reduces the parameter space to eight key groups, facilitating systematic exploration. Full derivations and numerical details are given in Appendices A–C.

II. MODEL FORMULATION

A. Physical system

We consider a thermoresponsive gel body containing an immobilized catalyst that catalyzes an exothermic, irreversible reaction consuming a substrate (reactant) supplied by the surrounding bath. Two geometries are studied: (i) a *symmetric slab* of dry-state half-thickness H_0, with Lagrangian coordinate $\xi = X/H_0 \in [0,1]$; and (ii) a *sphere* of dry-state radius R_0, with radial coordinate $s = R/R_0 \in [0,1]$. In both cases, $\xi = 0$ (or $s = 0$) is the symmetry center and $\xi = 1$ (or $s = 1$) is the free surface.

B. Governing equations

The three primary fields are the local swelling ratio J, the normalized reactant concentration u (scaled by the bath value), and the dimensionless temperature excess $\theta = (T - T_0)/\Delta T_{\mathrm{ref}}$. The dimensionless governing equations (derived in Appendix B) are

$$\frac{\partial J}{\partial t} = -\frac{\partial}{\partial \xi}\Big[\mathcal{M}(J)\,\frac{\partial \mu}{\partial \xi}\Big], \tag{1a}$$

$$\frac{\partial (uJ)}{\partial t} = \frac{\partial}{\partial \xi}\Big[\mathcal{D}(J)\,\frac{\partial u}{\partial \xi}\Big] - \mathrm{Da}\,J\,\widehat{R}(u,J,\theta), \tag{1b}$$

$$\mathcal{C}(J)\,\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial \xi}\Big[\mathcal{K}(J)\,\frac{\partial \theta}{\partial \xi}\Big] + \mathrm{Da}\,J\,\widehat{R}(u,J,\theta), \tag{1c}$$

where $\mathcal{M}$ is the swelling mobility, $\mathcal{D}$ and $\mathcal{K}$ are the effective reactant diffusivity and thermal conductivity, and $\mathcal{C}$ is the effective heat capacity. The reaction rate is

$$\widehat{R} = u\,(1-\phi)^{m_{\mathrm{act}}}\,\exp\!\Big(\frac{\Gamma_A\,\theta}{1+\varepsilon_T\,\theta}\Big), \tag{2}$$

with $\phi = \phi_{p0}/J$. The solvent chemical potential $\mu[J,\theta]$ is derived from the Flory–Rehner free energy with a temperature-dependent χ parameter exhibiting an LCST transition (Appendix A).

For the spherical geometry, the Laplacian in Eqs. (1) acquires the factor $(1/s^2)\partial_s(s^2\cdot)$, and the elastic contribution to μ includes anisotropic stretch corrections (Appendix A).

C. Boundary conditions

At the symmetry center ($\xi = 0$ or $s = 0$), all fluxes vanish. At the free surface ($\xi = 1$ or $s = 1$):

$$\mu\big|_{\xi=1} = 0, \quad \mathcal{D}\,\frac{\partial u}{\partial \xi}\Big|_{\xi=1} = \mathrm{Bi}_c\,(1-u), \quad \mathcal{K}\,\frac{\partial \theta}{\partial \xi}\Big|_{\xi=1} = -\mathrm{Bi}_T\,\theta. \tag{3}$$

D. Lumped (0D) model

The spatially uniform limit yields a three-ODE system for $J(t)$, $u(t)$, $\theta(t)$, where surface exchange enters through Biot-number terms:

$$\dot{J} = \mathrm{Bi}_\mu\,J^{2/3}\big[\mu_b - \mu(J,\theta)\big], \tag{4a}$$

$$\frac{d(Ju)}{dt} = \mathrm{Bi}_c\,J^{2/3}(1-u) - \mathrm{Da}\,J\,\widehat{R}, \tag{4b}$$

$$\mathcal{C}(J)\,\dot{\theta} = \mathrm{Da}\,J\,\widehat{R} - \mathrm{Bi}_T\,J^{2/3}\,\theta. \tag{4c}$$

This system is amenable to standard bifurcation analysis and provides analytical insight into the oscillation mechanism.

E. Dimensionless parameters

The eight control parameters are summarized in Table I.

TABLE I. Key dimensionless parameters.

Symbol	Physical meaning	Default	Range
Da	Reaction / diffusion rate	4	3–18
Bi_c	Surface reactant exchange	0.70	0.1–2
Bi_T	Surface heat loss	0.10	0.05–0.5
S_χ	LCST sensitivity	1.0	0.5–8
Γ_A	Arrhenius sensitivity	1.5	0–7
D_0	Reactant diffusivity scale	2.0	0.5–3
m_{act}	Reaction accessibility exponent	6	1–6
m_{diff}	Diffusion tortuosity exponent	2	1–6

III. LINEAR STABILITY OF THE 1D SLAB

Before turning to the fully nonlinear simulations, we analyze the stability of *uniform* slab states of the PDE system (1). This calculation supplies the analytical dispersion relation underlying Fig. 1 and clarifies which parameter combinations destabilize the homogeneous state through a long-wave Hopf mode and which instead amplify finite-wavenumber spatial structure.

A. Uniform base state and perturbation ansatz

We first seek spatially uniform steady states (J_0, u_0, θ_0), for which all gradients vanish. The boundary conditions then reduce to the algebraic balances

$$\mu(J_0,\theta_0) = \mu_b, \qquad \mathrm{Bi}_c(1-u_0) = \mathrm{Da}\,J_0\,\widehat{R}_0, \qquad \mathrm{Bi}_T\theta_0 = \mathrm{Da}\,J_0\,\widehat{R}_0, \tag{5}$$

where $\widehat{R}_0 = \widehat{R}(u_0, J_0, \theta_0)$. Thus the base state is set by chemical equilibrium together with the balance between

reaction, surface reactant supply, and heat loss. For the unit slab used here, the latter two equalities can be interpreted as volume-averaged Robin exchange conditions.

We perturb this base state by normal modes of the form

$$\begin{bmatrix} J \\ u \\ \theta \end{bmatrix} = \begin{bmatrix} J_0 \\ u_0 \\ \theta_0 \end{bmatrix} + \begin{bmatrix} \widehat{J} \\ \widehat{u} \\ \widehat{\theta} \end{bmatrix} e^{\sigma t + ik\xi}, \qquad k \geq 0, \tag{6}$$

where $\sigma(k)$ is the growth rate and k is the slab wavenumber. The long-wave mode $k = 0$ corresponds to a spatially uniform oscillation, whereas $k > 0$ probes finite-wavelength pattern-forming tendencies.

B. Linearization and dispersion matrix

It is convenient to linearize the mixed formulation (C1)–(C4), in which the chemical potential entering the swelling flux is written as

$$m = f(J, \theta) - \ell^2 J_{\xi\xi}, \tag{7}$$

with $f(J, \theta) = m_{\mathrm{mix}}(J, \theta) + m_{\mathrm{el}}(J)$. Writing

$$f_J = \left.\frac{\partial f}{\partial J}\right|_0, \qquad f_\theta = \left.\frac{\partial f}{\partial \theta}\right|_0, \tag{8a}$$

$$R_J = \left.\frac{\partial \widehat{R}}{\partial J}\right|_0, \qquad R_u = \left.\frac{\partial \widehat{R}}{\partial u}\right|_0, \tag{8b}$$

$$R_\theta = \left.\frac{\partial \widehat{R}}{\partial \theta}\right|_0, \tag{8c}$$

and evaluating the transport coefficients at the base state,

$$\mathcal{M}_0 = \mathcal{M}(J_0, \theta_0), \qquad \mathcal{D}_0 = \delta\, \mathcal{D}(J_0, \theta_0), \tag{9a}$$

$$\mathcal{K}_0 = \alpha\, \mathcal{K}(J_0), \qquad \mathcal{C}_0 = \mathcal{C}(J_0), \tag{9b}$$

the linearized problem can be written for $\widehat{\mathbf{y}} = (\widehat{J}, \widehat{u}, \widehat{\theta})^T$ as

$$\sigma\, \widehat{\mathbf{y}} = \mathbf{A}(k)\, \widehat{\mathbf{y}}, \qquad \mathbf{A}(k) = \mathbf{A}_0 + k^2 \mathbf{D}_2 + k^4 \mathbf{D}_4. \tag{10}$$

Here $\mathbf{A}_0$ contains the reaction and boundary-exchange terms,

$$\mathbf{A}_0 = \begin{bmatrix} a_{11} & 0 & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \tag{11}$$

with

$$a_{11} = -\mathrm{Bi}_\mu f_J, \qquad a_{13} = -\mathrm{Bi}_\mu f_\theta, \tag{12a}$$

$$a_{21} = \frac{u_0 \mathrm{Bi}_\mu f_J - \mathrm{Da}(\widehat{R}_0 + J_0 R_J)}{J_0}, \quad a_{22} = \frac{-\mathrm{Bi}_c - \mathrm{Da} J_0 R_u}{J_0}, \tag{12b}$$

$$a_{23} = \frac{u_0 \mathrm{Bi}_\mu f_\theta - \mathrm{Da} J_0 R_\theta}{J_0}, \qquad a_{31} = \frac{\mathrm{Da}(\widehat{R}_0 + J_0 R_J)}{\mathcal{C}_0}, \tag{12c}$$

$$a_{32} = \frac{\mathrm{Da} J_0 R_u}{\mathcal{C}_0}, \qquad a_{33} = \frac{-\mathrm{Bi}_T + \mathrm{Da} J_0 R_\theta}{\mathcal{C}_0}. \tag{12d}$$

the k^2 terms collect ordinary diffusion and thermo-chemo coupling,

$$\mathbf{D}_2 = \begin{bmatrix} -\mathcal{M}_0 f_J & 0 & -\mathcal{M}_0 f_\theta \\ 0 & -\mathcal{D}_0/J_0 & 0 \\ 0 & 0 & -\mathcal{K}_0/\mathcal{C}_0 \end{bmatrix}, \tag{13}$$

and the only k^4 contribution comes from the Cahn–Hilliard regularization in the swelling chemical potential,

$$\mathbf{D}_4 = \begin{bmatrix} -\mathcal{M}_0 \ell^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{14}$$

Equation (10) is the central analytical result: the uniform-state stability reduces to a 3×3 eigenvalue problem for each wavenumber k.

C. Classification criteria and interpretation

For each parameter set we solve (5) for all admissible uniform branches, evaluate the eigenvalues of $\mathbf{A}(k)$, and classify the leading mode. In panels **(a)**, **(b)**, **(d)**, **(e)**, and **(f)** of Fig. 1, the classification is based on the $k = 0$ mode:

- **Stable**: $\mathrm{Re}(\sigma_1(0)) < 0$.
- **Hopf unstable**: $\mathrm{Re}(\sigma_1(0)) > 0$ and $\mathrm{Im}(\sigma_1(0)) \neq 0$.
- **Monotone unstable**: $\mathrm{Re}(\sigma_1(0)) > 0$ and $\mathrm{Im}(\sigma_1(0)) = 0$.

Panel **(c)** additionally shows the full dispersion curve $\max_j \mathrm{Re}(\sigma_j(k))$ and therefore exposes finite-k growth that is invisible in a purely lumped analysis. When $f_J < 0$, the homogeneous state lies inside the spinodal region, and the competition between the destabilizing k^2 term in (13) and the stabilizing $-\mathcal{M}_0 \ell^2 k^4$ term in (14) produces a finite-wavenumber maximum at k^*. The inset of Fig. 1(c) isolates the long-wave Hopf behavior near $k = 0$, while the main panel highlights this short-wave spinodal peak.

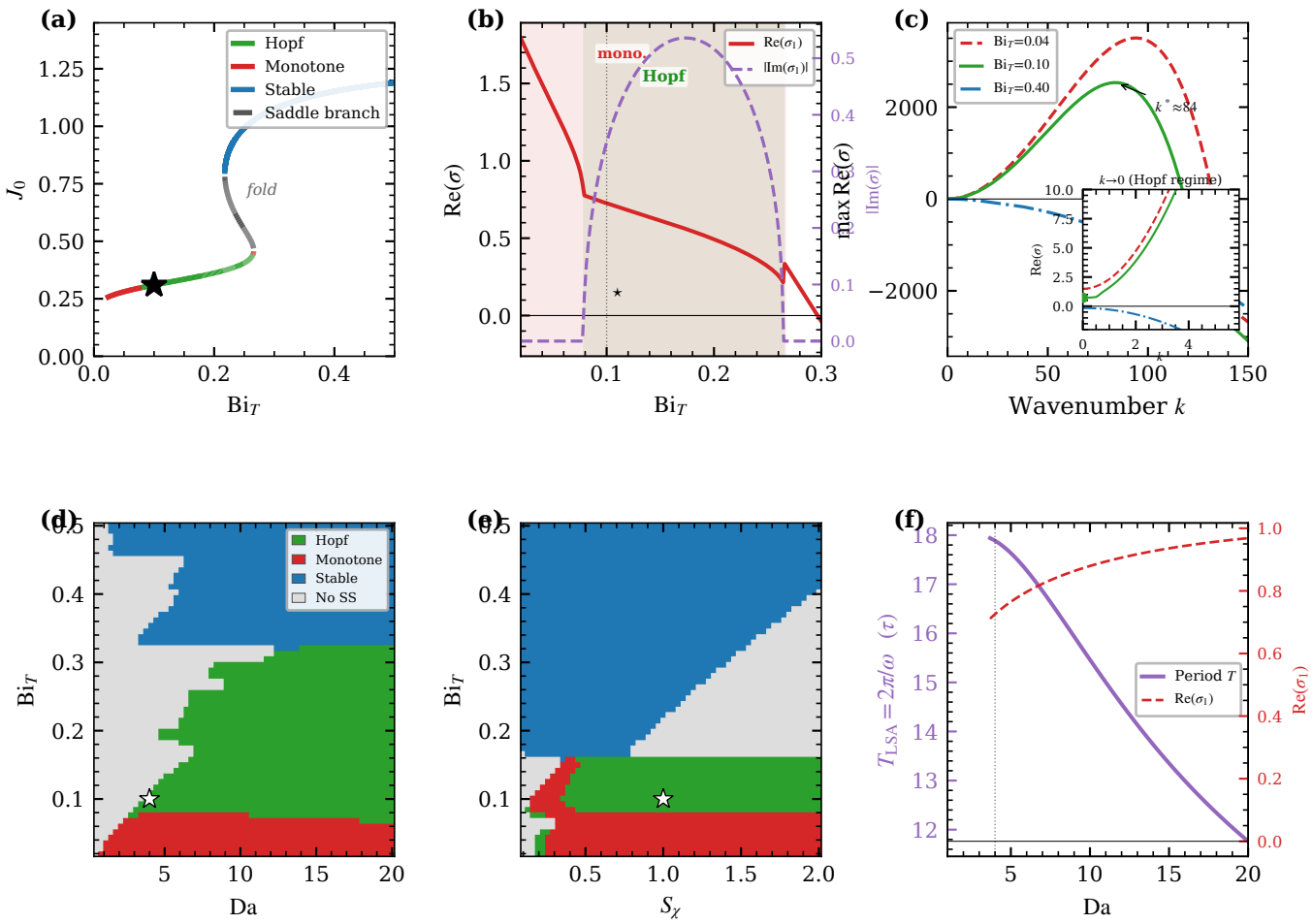

FIG. 1. Linear stability of the 1D slab based on the analytical dispersion relation. **(a)** Steady-state branches J_0 versus Bi_T, colored by the stability of the $k = 0$ mode. Stable segments are drawn solid, unstable segments dashed, and the middle saddle branch is shown in gray; the star marks the working point used in the nonlinear simulations. **(b)** Real and imaginary parts of the leading $k = 0$ eigenvalue along the collapsed branch, showing the monotone-to-Hopf transition and the finite Bi_T oscillation window. **(c)** Dispersion curves $\max \mathrm{Re}(\sigma(k))$ for three representative Bi_T values. The inset resolves the long-wave behavior near $k = 0$, while the main panel shows the finite-k spinodal peak. **(d)** Classification map in the $(\mathrm{Da}, \mathrm{Bi}_T)$ plane: stable (blue), Hopf unstable (green), monotone unstable (red), and no admissible uniform steady state (gray). **(e)** Corresponding classification map in the (S_χ, Bi_T) plane. **(f)** Linear period estimate $T_{\mathrm{LSA}} = 2\pi/\omega$ and growth rate $\mathrm{Re}(\sigma_1)$ versus Da at the working point.

D. Implications for the working point

At the default parameter set used for the slab simulations, the relevant uniform state lies on the collapsed branch in Fig. 1(a). Panels **(a)** and **(b)** show that increasing Bi_T drives a sequence monotone $\to$ Hopf $\to$ stable/fold: weak cooling causes thermal runaway, intermediate cooling yields oscillatory growth, and sufficiently strong cooling stabilizes the swollen branch or removes the collapsed branch altogether. Panels **(d)** and **(e)** show that this Hopf region persists only when the reaction strength (Da) and LCST sensitivity (S_χ) are large enough to overcome surface heat loss.

The analytical period estimate in Fig. 1(f), $T_{\mathrm{LSA}} = 2\pi/\omega$, provides the small-amplitude precursor to the nonlinear oscillation period measured in the direct PDE simulations below. Quantitative differences are expected once the orbit reaches finite amplitude, but the dispersion analysis correctly identifies the existence, type, and timescale of the instability that seeds the self-oscillating slab dynamics.

IV. SPATIALLY RESOLVED DYNAMICS IN THE 1D SLAB

We now turn to the full 1D slab model, which resolves spatial gradients that are absent in the 0D reduction.

A. Self-sustained oscillation mechanism

Figure 2 shows the oscillation dynamics at the default parameters (Table I). After a brief transient ($\sim 3\tau$), the system settles into a limit cycle with period $T \approx 27\tau$ and surface amplitude $\Delta J \approx 0.93$. The oscillation follows a relaxation cycle with four distinct phases (Fig. 2(c)):

1. **Swelling** (J rising, $\theta \approx 0$): the gel is below the LCST and absorbs solvent; fresh reactant permeates the swollen network.
2. **Ignition** (θ rising steeply): accumulated reactant drives exothermic heat release; θ crosses the LCST threshold.
3. **Collapse** (J dropping, θ peaked): the gel deswells sharply, expelling solvent and reactant; pore accessibility drops to near zero.
4. **Cooling / refill** (θ falling, J at minimum): heat dissipates through the surface while reactant access recovers; once $\theta < \theta_{\mathrm{LCST}}$, swelling restarts.

The phase portrait (Fig. 2(d)) shows a well-defined limit cycle in the $(\langle J\rangle, \langle\theta\rangle)$ plane, confirming the relaxation oscillator character.

B. Dual-zone termination mechanism

The spatial structure reveals a key feature absent from the 0D model: the oscillation terminates through different mechanisms at different locations. In the *outer shell* ($\xi \gtrsim 0.7$), the gel collapses sharply and the accessibility factor $(1-\phi)^{m_{\mathrm{act}}} \to 0$, extinguishing the reaction even though the local reactant concentration remains non-negligible ($u_{\mathrm{surf}} \approx 0.23$–$0.28$ at the collapse moment). This is *accessibility-dominated* termination: the mechanical collapse, not reactant depletion, shuts off the heat source. After collapse, u_{surf} rises to ≈ 1 as the surface Robin supply continues while the zero-accessibility shell blocks consumption; this "stored" reactant re-ignites the next cycle.

In the *inner core* ($\xi \lesssim 0.5$), the Thiele modulus $\Phi^2 = \mathrm{Da}\,(1-\phi_0)^{m_{\mathrm{act}}}/(\delta D_0)$ is large enough that u is depleted during the heating phase while the core remains swollen. Here termination is *reactant-depletion dominated*. This spatial coexistence of accessibility-controlled and depletion-controlled termination is a hallmark of the thermo-chemo-poroelastic coupling and has no analog in spatially uniform oscillator models.

Figure 3 resolves this cycle directly in space and time. The surface traces in panels **(a)**–**(c)** show that collapse is triggered while u_{surf} remains finite, whereas the kymographs and phase-selected profiles in panels **(d)**–**(g)** show that accessibility loss is localized near the boundary and that the inner core remains reactant limited. The mechanism is therefore genuinely dual-zone: the outer shell is accessibility-terminated, while the interior is depletion-limited.

C. Two oscillatory regimes

Consistent with the linear-stability predictions (Sec. III), the 1D slab exhibits two qualitatively distinct oscillatory regimes (Fig. 4):

Regime I ($S_\chi \lesssim 0.8$): Large-amplitude relaxation oscillations ($\Delta J \gtrsim 0.4$) with long period ($T \gtrsim 30\tau$). Temperature excursions are large ($\Delta\theta \sim 2$–3), and the spatial structure is strongly nonuniform: the surface collapses to $J \approx 0.15$ while the center remains near equilibrium (Fig. 4(b,d,f)).

Regime II ($S_\chi \gtrsim 1.0$): Small-amplitude quasi-harmonic oscillations ($\Delta J \lesssim 0.1$, $T \lesssim 15\tau$). The gel barely crosses the LCST and the spatial profiles are more uniform (Fig. 4(a,c,e)).

The kymographs (Fig. 4(c,d)) highlight the spatial extent of the oscillation: in Regime I, only the outer $\sim$30% of the slab participates in the collapse–swelling cycle; in Regime II, the entire slab oscillates nearly uniformly.

D. Parameter-space phase diagram

Figure 5 presents the oscillation landscape in the (S_χ, Da) plane. The phase diagram (Fig. 5(a)) distinguishes oscillatory from steady-state regions and identifies the boundary between Regimes I and II. The corresponding bifurcation curves (Fig. 5(b)) show that the oscillation amplitude grows monotonically with Da in Regime I but remains bounded in Regime II.

The linear-stability maps in Fig. 1(d,e) organize the nonlinear oscillation window in the full slab simulations: oscillation appears only inside the finite Hopf band, while low Bi_T leads to monotone thermal runaway and large Bi_T stabilizes the swollen branch or eliminates the collapsed uniform state altogether. The nonlinear 1D dynamics of Fig. 5 therefore emerge from this same stability structure, with quantitative shifts caused by finite-amplitude transport delay and spatially nonuniform evolution.

The ratio $\mathrm{Bi}_c/\mathrm{Bi}_T$ governs the competition between reactant supply and heat loss. Oscillation requires $1.5 \lesssim \mathrm{Bi}_c/\mathrm{Bi}_T \lesssim 15$; outside this window, either insufficient reactant supply or excessive cooling prevents the thermal runaway from completing the feedback loop. Figure 5(c) shows that increasing Bi_c monotonically increases the reactant penetration depth, providing a route to enhance interior reactant delivery.

V. GEOMETRY EFFECTS: SLAB VS. SPHERE

The instability and oscillation mechanism identified in Secs. III–IV relies on the interplay between surface-driven heat loss, reactant supply, and the gel's volume-collapse response. Geometry affects each of these processes through the surface-to-volume ratio and the form

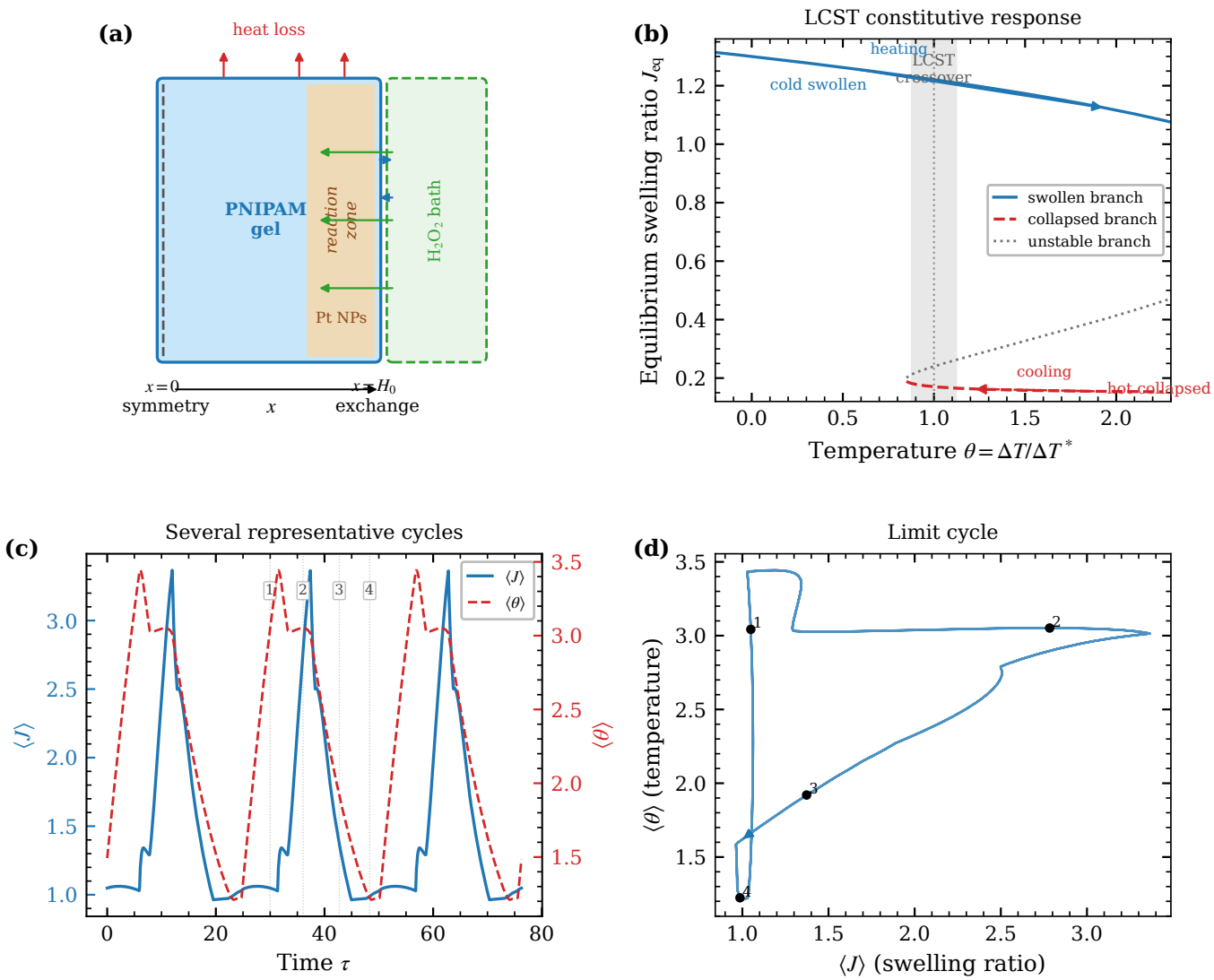

FIG. 2. Self-sustained oscillation mechanism in the 1D slab. **(a)** Gel slab–bath schematic showing the three coupled fields and boundary conditions. **(b)** Representative LCST constitutive response: equilibrium swelling ratio J_{eq} versus temperature θ, showing the swollen, collapsed, and unstable branches, together with schematic heating and cooling trajectories across the LCST crossover region. **(c)** Time series of $\langle J\rangle$ (blue, left axis) and $\langle\theta\rangle$ (red, right axis) over several representative cycles; numbered guides mark the four stages: swelling, ignition, collapse, and cooling/refill. **(d)** Phase portrait showing the limit cycle in the $(\langle J\rangle, \langle\theta\rangle)$ plane, with numbered points corresponding to the four stages in panel (c).

of the divergence operator. We now compare the 1D slab with the 1D spherically symmetric geometry.

A. Isotropic sphere model

We first consider a spherical gel bead with the same material parameters as the slab but with the isotropic elastic energy $\mu_{\mathrm{el}} = \Omega_e(J^{-1/3} - J^{-1})$, neglecting the distinction between radial and circumferential stretches. The governing equations are identical to Eqs. (1) except that the spatial operators acquire the spherical factor $(1/s^2)\partial_s(s^2\cdot)$.

B. Critical diffusion coupling

To probe the effect of internal transport, we introduce a diffusion coupling parameter $\varepsilon_{\mathrm{diff}}$ that continuously interpolates between a surface-only model ($\varepsilon_{\mathrm{diff}} = 0$, where only the outermost shell exchanges with the bath) and the fully coupled model ($\varepsilon_{\mathrm{diff}} = 1$). Figure 6(b) shows the oscillation amplitude $\Delta\theta$ as a function of $\varepsilon_{\mathrm{diff}}$:

- At $\varepsilon_{\mathrm{diff}} = 0$, the sphere oscillates with dynamics qualitatively similar to the slab (Fig. 6(d)).
- Above a critical threshold $\varepsilon^* \approx 3 \times 10^{-4}$, the oscillation amplitude drops to zero and the system settles into a steady swollen state (Fig. 6(e)).

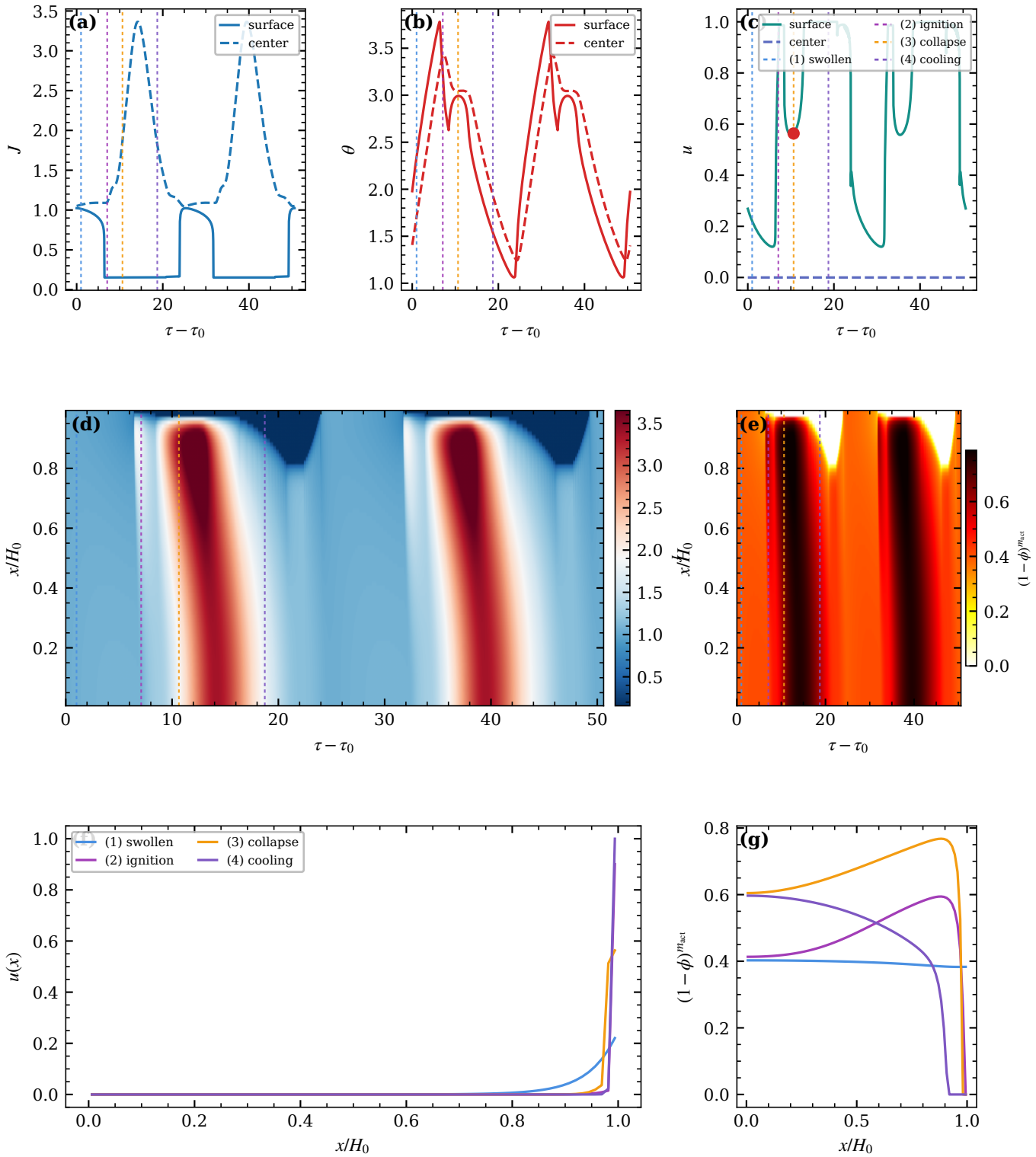

FIG. 3. Detailed spatiotemporal evidence for the oscillation mechanism at the optimized slab parameters (Da = 4, $D_0 = 2$, $\mathrm{Bi}_c = 0.70$, $\mathrm{Bi}_T = 0.10$). **(a)** Surface and center swelling-ratio traces. **(b)** Surface and center temperature traces. **(c)** Surface and center reactant traces; the marked collapse event occurs at finite u_{surf}. **(d)** Spatiotemporal kymograph of $J(x, \tau)$. **(e)** Kymograph of the accessibility factor $(1-\phi)^{m_{\mathrm{act}}}$. **(f)** Reactant profiles at the four representative phases. **(g)** Accessibility profiles at the same phases.

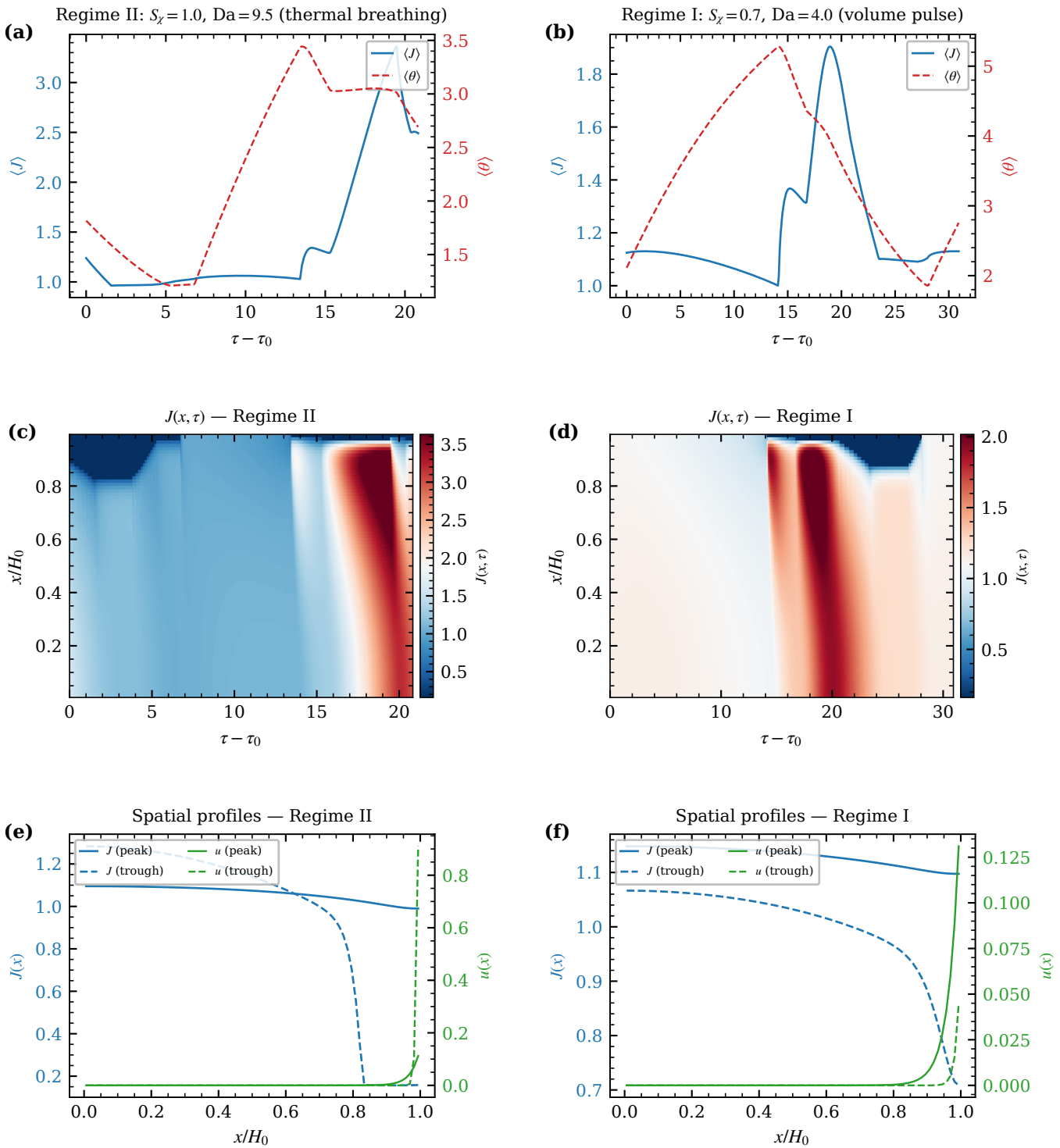

FIG. 4. Two oscillatory regimes in the 1D slab. *Left column*: Regime II (Da = 9.5, $S_\chi = 1.0$); *right column*: Regime I (Da = 4.0, $S_\chi = 0.7$). **(a,b)** Time series of $\langle J \rangle$ (left axis) and $\langle \theta \rangle$ (right axis) over the last three periods. **(c,d)** Kymographs of $J(\xi, t)$ showing the spatiotemporal structure. **(e,f)** Spatial profiles of J (blue), θ (red), and u (green) at peak (solid) and trough (dashed) of the oscillation.

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FIG. 5. Parameter-space structure of the 1D slab. **(a)** $S_\chi \times$ Da phase diagram (16×16 grid, $N = 51$, $t_{\rm end} = 400\tau$): colors indicate oscillatory vs steady-state classifications; white contours show ΔJ levels. **(b)** Oscillation amplitude ΔJ (solid) and $\Delta\theta$ (dashed) vs Da for Regime I ($S_\chi = 0.7$, red) and Regime II ($S_\chi = 1.0$, blue). **(c)** Reactant penetration depth ℓ_u/H_0 vs Bi_c at Da $= 9.5$, $S_\chi = 1.0$.

The oscillation-killing mechanism is geometric: in the sphere, the surface-to-volume ratio $3/R$ is larger than the slab's $1/H_0$, and solvent can permeate the gel from all directions. The enhanced radial influx creates an expanding swelling wavefront that outpaces the collapse domain. Once the wavefront reaches the center, the entire gel is swollen before the thermal runaway can complete, and the oscillation dies.

C. Anisotropic stress effects

In a spherical geometry, radial and circumferential stretches differ ($\lambda_r \neq \lambda_\theta$), generating anisotropic stress. The full-mechanics model reconstructs the current radius $r^3(R) = 3\int_0^R J\,R'^2\,dR'$, computes $\lambda_r = J/\lambda_\theta^2$, and solves the stress equilibrium $\partial\sigma_r/\partial r + (2/r)(\sigma_r - \sigma_\theta) = 0$ to obtain a nonlocal radial stress correction $\Sigma_r(R)$ that enters the chemical potential.

This stress locking creates a core–shell quasi-equilibrium: the interior remains at $J \approx 0.7$, and the anisotropic stress prevents the radial relaxation needed to complete the swelling–collapse cycle. As a result, the full-mechanics sphere does not oscillate at normal diffusion coupling, even in parameter regimes where the slab and 0D models predict robust oscillation.

D. Physical implications

The geometry comparison reveals that the slab is the most favorable geometry for thermal oscillation:

- **Slab**: single exposed surface → limited solvent influx → collapse is effective → oscillation is robust.
- **Sphere**: all-round exposure → rapid radial influx → swelling wavefront escapes collapse domain → oscillation suppressed.

This finding provides direct guidance for experimental design: flat membranes or thin films should be preferred over spherical beads for realizing thermally driven gel oscillators.

VI. REACTANT PENETRATION ENHANCEMENT

For applications such as periodic chemical delivery or pulsatile drug release, it is essential that the reactant pen-

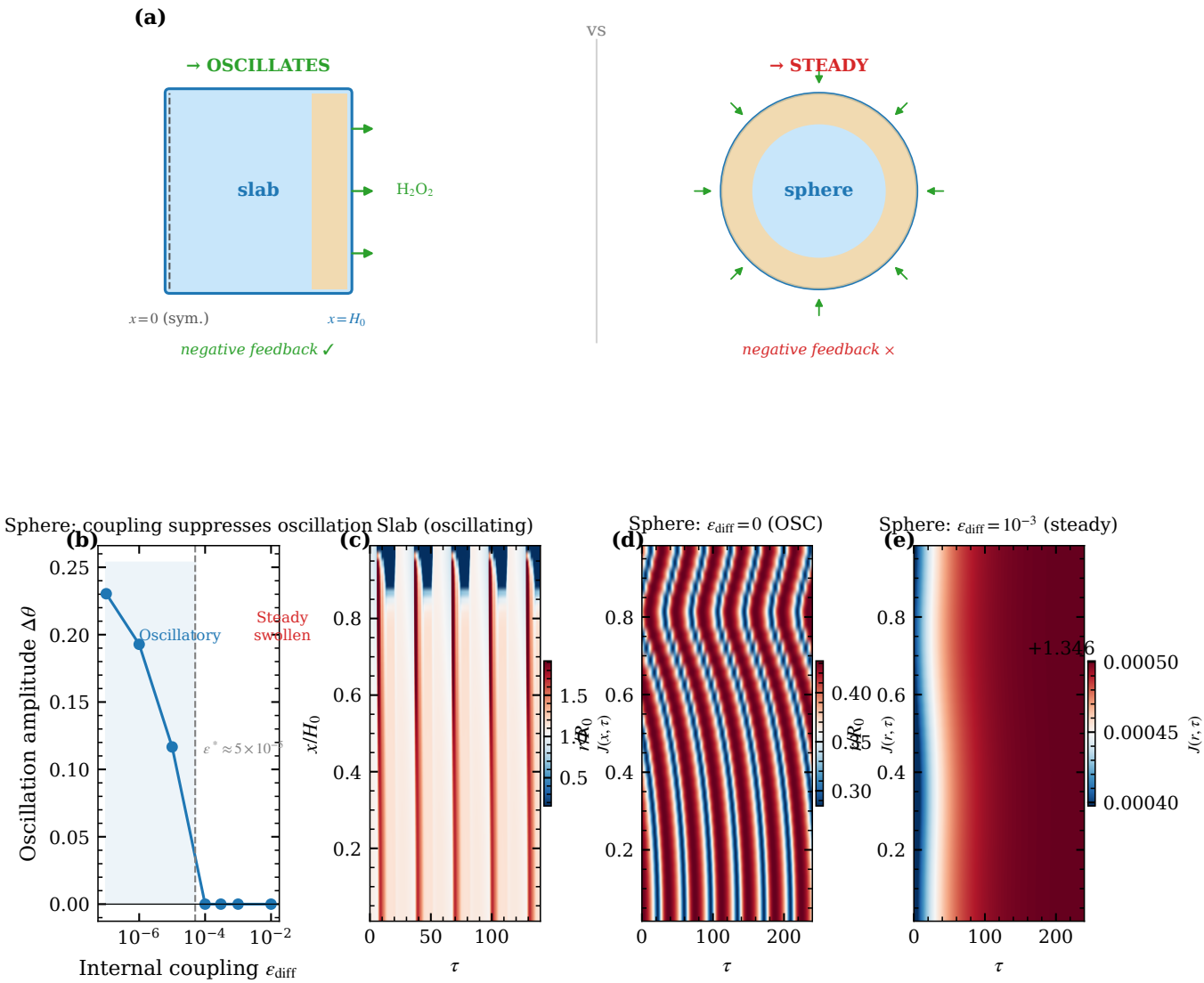

FIG. 6. Geometry effects on oscillation. **(a)** Schematics of slab (left) and sphere (right) geometries with boundary conditions. **(b)** Oscillation amplitude $\Delta\theta$ vs diffusion coupling $\varepsilon_{\rm diff}$ in the sphere; the critical threshold $\varepsilon^* \approx 3\times10^{-4}$ is marked. **(c)** Slab kymograph $J(\xi,t)$ at Regime I parameters (reference case, oscillates). **(d)** Sphere kymograph at $\varepsilon_{\rm diff}=0$ (surface-only model, oscillates). **(e)** Sphere kymograph at $\varepsilon_{\rm diff}=10^{-3}$ (fully coupled, steady state).

etrate deep into the gel interior. We now characterize the penetration depth and identify optimization strategies.

A. Accessibility gradient and dual-regime structure

The interior reactant concentration depends on the competition between diffusion supply and reaction consumption, quantified by the Thiele modulus $\Phi^2 = \mathrm{Da}\,(1-\phi_0)^{m_{\rm act}}/(\delta D_0)$. Figure 8(a) shows the center reactant concentration $u_{\rm ctr}$ as a function of Da: at high Da, the interior is completely depleted ($u_{\rm ctr}\to 0$), while at low Da, significant reactant persists throughout the cycle.

The accessibility front depth x^*—the fractional depth at which $u > 0.05$ at collapse onset—follows the Thiele scaling (Fig. 8(b)):

$$x^* \sim \Phi^{-1} = \sqrt{\frac{\delta D_0}{\mathrm{Da}\cdot\overline{\alpha}_{\rm h}\cdot\overline{f}_{\rm h}}}, \tag{15}$$

where $\overline{\alpha}_{\rm h}$ and $\overline{f}_{\rm h}$ are the accessibility and Arrhenius factors at mid-heating conditions.

B. Da–Γ_A mechanism map

Figure 8(c) maps the oscillation mechanism in the (Da,Γ_A) plane. Two regimes emerge:

1. **Depletion-dominated** (large Da, small Γ_A): the inner core is starved of reactant; termination oc-

curs by depletion in the interior and accessibility collapse at the surface.

2. **Accessibility-dominated** (small Da, large Γ_A): reactant persists throughout the gel; termination everywhere occurs via the pore-closure mechanism.

The boundary satisfies

$$\mathrm{Da}^* = \frac{\delta D_0}{(1-\phi_0/J_{\mathrm{ctr}})^{m_{\mathrm{act}}} \exp\bigl(\Gamma_A \theta_h/(1+\theta_h)\bigr)}, \qquad (16)$$

evaluated at $\theta_h \approx 1.5$ and $J_{\mathrm{ctr}} \approx 1.15$.

C. Optimization strategy

We identify a three-parameter strategy to maximize interior penetration while maintaining oscillation:

1. **Reduce** Da: slower reaction allows diffusion to carry u deeper. Reducing from $\mathrm{Da} = 14$ to $\mathrm{Da} = 4$ increases u_{mid} by $\sim 100\times$.

2. **Increase** D_0: higher diffusivity accelerates interior supply. $D_0 = 2$ gives full penetration, but $D_0 > 2.5$ homogenizes u and destroys the spatial feedback needed for oscillation.

3. **Increase** Bi_c: stronger surface supply compensates for the lower Da.

At the optimized parameters ($\mathrm{Da} = 4$, $D_0 = 2$, $\mathrm{Bi}_c = 0.70$, $\mathrm{Bi}_T = 0.10$), the mid-slab reactant concentration reaches $u_{\mathrm{mid}}^{\mathrm{peak}} = 4.5 \times 10^{-3}$—a **590**$\times$ enhancement over the original parameter set ($\mathrm{Da} = 14$, $D_0 = 1$, $\mathrm{Bi}_c = 0.35$: $u_{\mathrm{mid}}^{\mathrm{peak}} \approx 7.6 \times 10^{-6}$)—while preserving $\Delta J \approx 0.93$ and achieving full-domain penetration (Fig. 7).

VII. DISCUSSION

A. Model hierarchy: 0D to 1D validation

The 0D lumped model successfully predicts the Hopf bifurcation boundaries and the existence of two oscillatory regimes, validating its use as a rapid screening tool for parameter exploration. The 1D slab confirms these predictions while adding crucial spatial physics: the dual-zone termination mechanism, nonuniform oscillation profiles, and the surface-reactant storage effect. Quantitatively, the 0D and 1D critical Da values agree to within $\sim 20\%$, with discrepancies attributable to the transport delay introduced by finite diffusion lengths.

B. Geometry selection for experiments

Our geometry comparison provides clear experimental guidance: flat membranes or thin films are strongly preferred over spherical beads for realizing thermally driven oscillators. The sphere's enhanced surface-to-volume ratio and anisotropic stress effects conspire to suppress the collapse–swelling cycle. This finding may explain why Siegel's original spherical-bead experiments [5] required carefully tuned conditions, while slab-geometry demonstrations [7] were more robust.

C. Application to chemical delivery

The dual-zone termination mechanism has implications for autonomous chemical delivery. The "stored reactant" phase ($u_{\mathrm{surf}} \approx 1$ during the collapsed state) provides a built-in reservoir that ensures reliable re-ignition, making the oscillation self-sustaining. The $590\times$ penetration enhancement demonstrates that the pulsatile pumping mechanism can deliver reactant deep into the gel interior, provided the Da–D_0–Bi_c combination is properly tuned. The counter-intuitive role of the accessibility exponent m_{act} deserves emphasis: higher m_{act} (sharper pore blockage) increases u_{mid} because it amplifies the J oscillation, driving stronger pulsatile injection during the swelling phase.

D. Limitations

The present model is one-dimensional and assumes uniaxial (slab) or isotropic (sphere) swelling; in three dimensions, shape changes and lateral stress could couple to the oscillation frequency. We neglect convective heat transfer ($\mathrm{Pe}_T = 0$) and mechanically mediated reactant transport. These are small corrections at the parameters studied (Appendix C). The model also assumes a single irreversible reaction; multi-step kinetics (as in the BZ reaction) may introduce additional dynamic complexity including period-doubling and chaos, which we leave to future work.

VIII. CONCLUSION

We have developed and analyzed a hierarchy of models—from analytical linear stability to spatially resolved 1D slab and sphere dynamics—for self-oscillating thermoresponsive gels driven by an exothermic reaction. The key contributions are:

1. A *systematic dimensionless framework* reducing the parameter space to eight groups with clear physical interpretations, enabling efficient exploration of the oscillation landscape.

2. *0D Hopf bifurcation analysis* identifying the critical conditions for oscillation onset in the (Da, S_χ) plane and predicting two oscillatory regimes: a large-amplitude relaxation oscillator (Regime I) and a small-amplitude quasi-harmonic oscillator (Regime II).

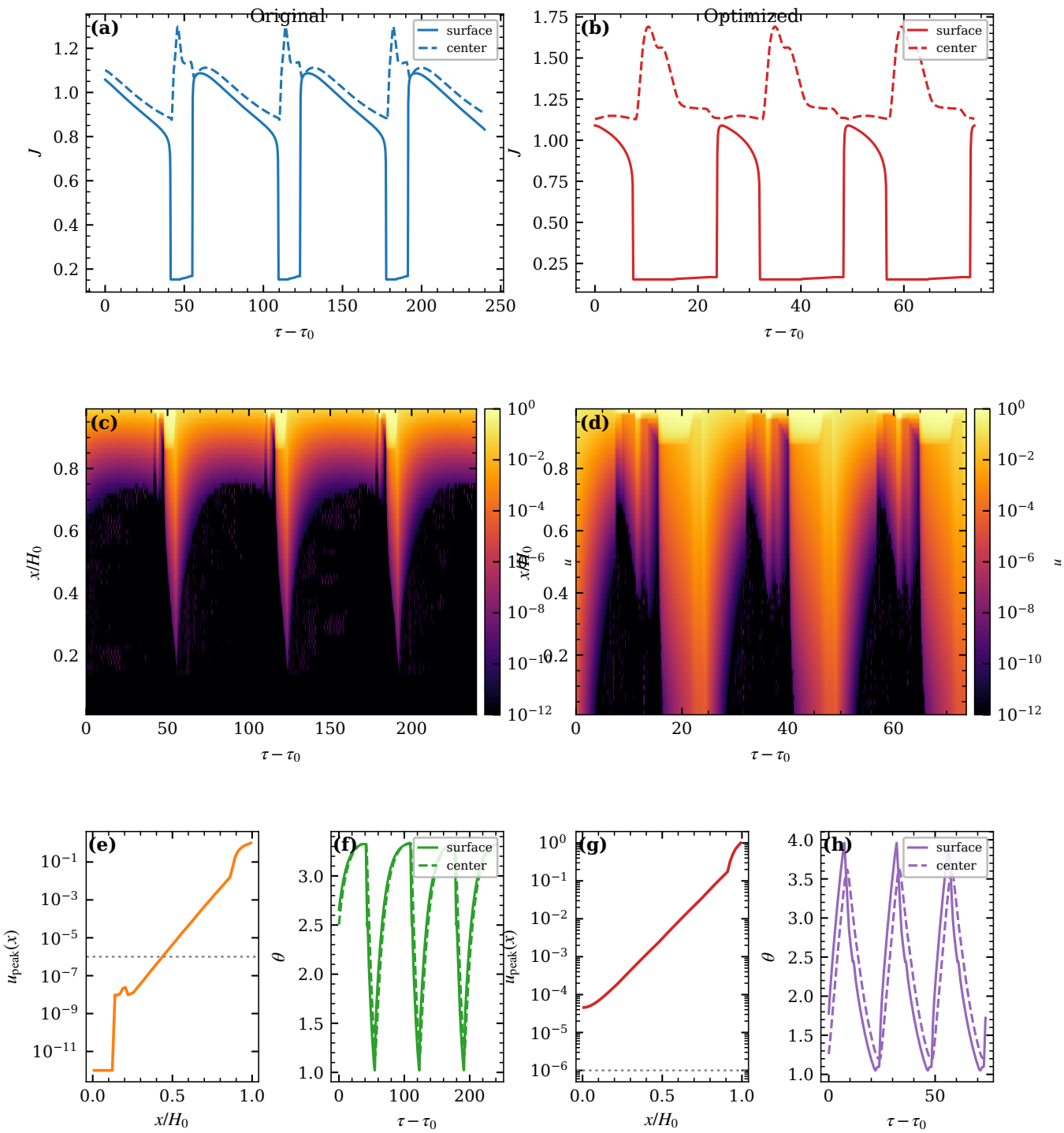

FIG. 7. Original-versus-optimized penetration dynamics. Left column pair: original parameters (Da = 14, $D_0 = 1$, $\mathrm{Bi}_c = 0.35$). Right column pair: optimized parameters (Da = 4, $D_0 = 2$, $\mathrm{Bi}_c = 0.70$). **(a,b)** Surface and center swelling-ratio traces. **(c,d)** Reactant kymographs $u(x, \tau)$. **(e,g)** Peak reactant profiles $u_{\mathrm{peak}}(x)$, showing the large increase in interior penetration after optimization. **(f,h)** Surface and center temperature traces, confirming that the optimized state remains oscillatory while substantially improving interior reactant delivery.

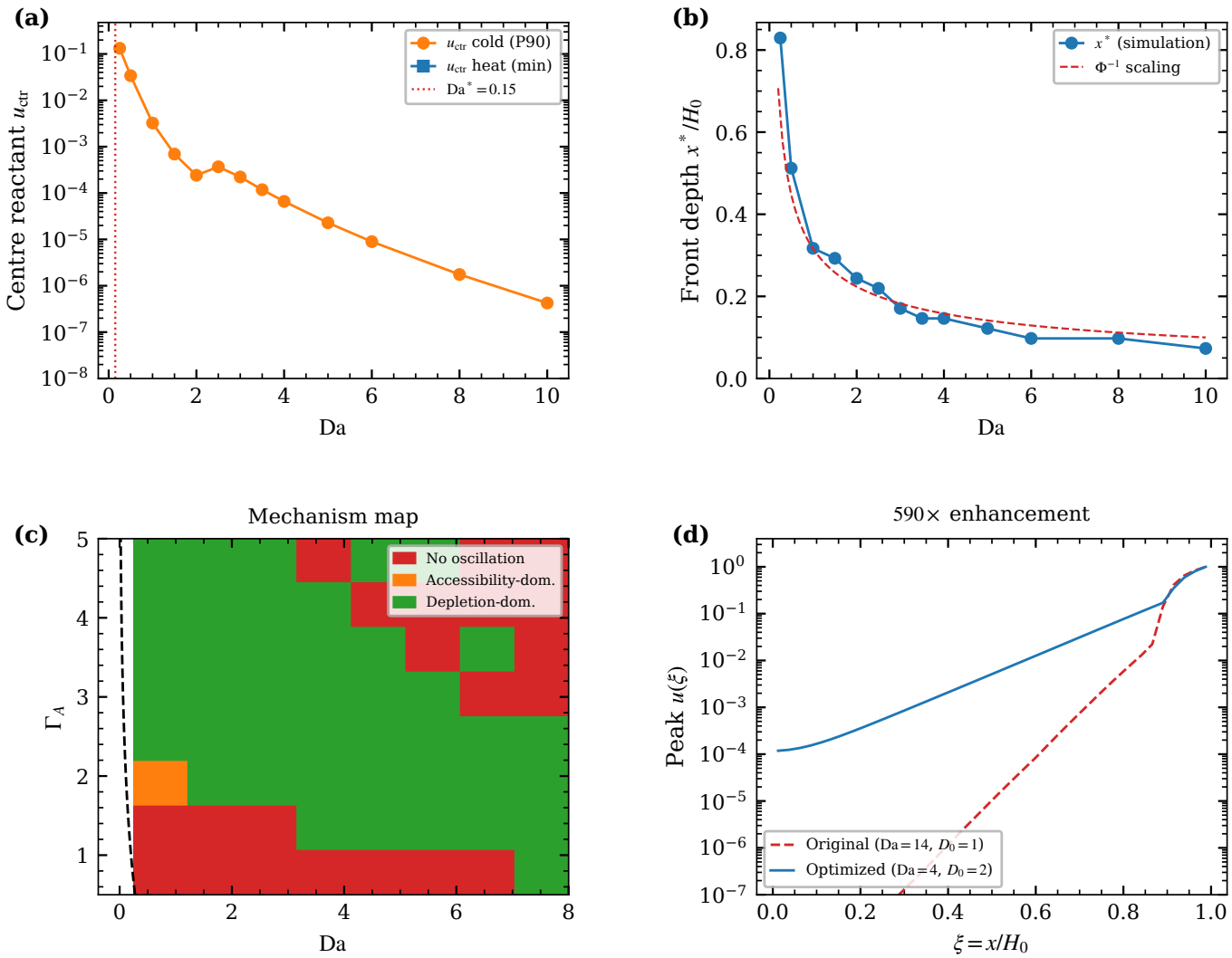

FIG. 8. Reactant penetration analysis and optimization. **(a)** Centre reactant u_{ctr} vs Da: cold-phase value (orange, 90th percentile) and heating-phase minimum (blue); red dotted line marks analytical Da^*. **(b)** Accessibility front depth x^* vs Da (circles), with analytical Φ^{-1} scaling (red dashed). **(c)** $\mathrm{Da} \times \Gamma_A$ mechanism map: yellow = accessibility-dominated, green = depletion-dominated, red = no oscillation; dashed curve = analytical $\mathrm{Da}^*(\Gamma_A)$ boundary. **(d)** Spatial profiles of $u(\xi)$ comparing the original ($\mathrm{Da} = 14$, $D_0 = 1$) and optimized ($\mathrm{Da} = 4$, $D_0 = 2$) parameters, demonstrating 590× enhancement of interior reactant level.

3. *Spatially resolved 1D slab dynamics* revealing the four-phase relaxation cycle, the dual-zone termination mechanism (accessibility-dominated outer shell vs. depletion-dominated inner core), and surface reactant storage enabling reliable re-ignition.

4. *Geometry-dependent oscillation*: the slab sustains robust oscillations, whereas the sphere suppresses them due to enhanced radial solvent influx and anisotropic stress locking. The critical diffusion coupling threshold is $\varepsilon^* \approx 3 \times 10^{-4}$.

5. *Reactant penetration optimization*: a 590× enhancement of interior reactant concentration is achieved by simultaneously reducing Da and increasing D_0 and Bi_c, while maintaining oscillation. The Thiele-modulus scaling provides a predictive design rule.

Future work will extend the model to two and three dimensions, map the full bifurcation diagram including period-doubling routes, and compare quantitatively with experimental observations on PNIPAM systems coupled to exothermic reactions.

ACKNOWLEDGMENTS

[Acknowledgments placeholder.]

[1] R. Yoshida and T. Ueki, NPG Asia Mater. **6**, e107 (2014).
[2] R. Yoshida, T. Takahashi, T. Yamaguchi, and H. Ichijo, J. Am. Chem. Soc. **118**, 5134 (1996).
[3] R. Yoshida, Adv. Mater. **22**, 3463 (2010).
[4] V. V. Yashin and A. C. Balazs, Science **314**, 798 (2006).
[5] R. A. Siegel and B. A. Firestone, Macromolecules **21**, 3254 (1988).
[6] R. A. Siegel, Adv. Polym. Sci. **109**, 233 (1993).
[7] R. Zarzycki, J. Controlled Release **317**, 1 (2020).
[8] J. Boissonade, Phys. Rev. Lett. **90**, 188302 (2003).
[9] P. J. Flory, *Principles of Polymer Chemistry* (Cornell University Press, Ithaca, 1953).
[10] T. Tanaka, Phys. Rev. Lett. **40**, 820 (1978).
[11] T. Tanaka and D. J. Fillmore, J. Chem. Phys. **70**, 1214 (1979).
[12] M. Heskins and J. E. Guillet, J. Macromol. Sci. A **2**, 1441 (1968).
[13] S. Hirotsu, Y. Hirokawa, and T. Tanaka, J. Chem. Phys. **87**, 1392 (1987).
[14] H. G. Schild, Prog. Polym. Sci. **17**, 163 (1992).
[15] M. Shibayama and T. Tanaka, Adv. Polym. Sci. **109**, 1 (1993).
[16] M. Quesada-Pérez, J. A. Maroto-Centeno, J. Forcada, and R. Hidalgo-Alvarez, Soft Matter **7**, 10536 (2011).
[17] M. Doi, J. Phys. Soc. Jpn. **78**, 052001 (2009).
[18] W. Hong, X. Zhao, J. Zhou, and Z. Suo, J. Mech. Phys. Solids **56**, 1779 (2008).
[19] S. A. Chester and L. Anand, J. Mech. Phys. Solids **58**, 1879 (2010).
[20] A. Lucantonio, P. Nardinocchi, and L. Teresi, J. Mech. Phys. Solids **61**, 205 (2013).
[21] O. Kuksenok and A. C. Balazs, Adv. Funct. Mater. **23**, 4601 (2013).
[22] P. Dayal, O. Kuksenok, and A. C. Balazs, Soft Matter **8**, 3814 (2012).
[23] A. M. Turing, Philos. Trans. R. Soc. B **237**, 37 (1952).
[24] M. Karg, I. Pastoriza-Santos, L. M. Liz-Marzán, and T. Hellweg, ChemPhysChem **7**, 2298 (2006).

Appendix A: Model Construction

We formulate a one-dimensional Lagrangian model for the coupled thermo-chemo-poroelastic dynamics of a thermoresponsive gel slab undergoing an exothermic reaction. The primary variables are the local swelling ratio $J(X,t)$, reactant concentration $c(X,t)$, and temperature $T(X,t)$.

1. Kinematics

Consider a gel slab of reference (dry-state) thickness H_0. We adopt a material (Lagrangian) coordinate $X \in [0, H_0]$, with $X = 0$ denoting the symmetry plane and $X = H_0$ the free surface. The current position of a material point is $x(X,t)$, and the local volume ratio is

$$J(X,t) = \frac{\partial x}{\partial X} > 0. \tag{A1}$$

In one dimension, J is the local swelling ratio. The polymer volume fraction is a dependent variable determined by mass conservation of the network:

$$\phi(X,t) = \frac{\phi_{p0}}{J(X,t)}, \tag{A2}$$

where ϕ_{p0} is the polymer volume fraction in the reference state. The current thickness of the slab is

$$h(t) = x(H_0,t) = \int_0^{H_0} J(X,t)\, dX. \tag{A3}$$

2. Governing equations

a. *Swelling–permeation equation*

The evolution of J is governed by solvent conservation:

$$\frac{\partial J}{\partial t} = -\frac{\partial Q_s}{\partial X}, \tag{A4}$$

where Q_s is the solvent flux relative to the polymer network, driven by chemical-potential gradients:

$$Q_s = -M_s(J,T)\,\frac{\partial \mu_s}{\partial X}. \tag{A5}$$

Here $M_s(J,T)$ is the permeation mobility and μ_s is the solvent chemical potential.

b. *Reactant transport equation*

The reactant concentration $c(X,t)$ (moles per current volume of pore fluid) evolves according to

$$\frac{\partial (Jc)}{\partial t} + \frac{\partial N_c}{\partial X} = -J\,R(c,T,J), \tag{A6}$$

with the total reactant flux

$$N_c = c\,Q_s - D_c(J,T)\,\frac{\partial c}{\partial X}. \tag{A7}$$

The first term represents advective transport by the solvent flow, and the second term Fickian diffusion within the pore fluid. The effective diffusivity $D_c(J,T)$ depends on the local microstructure via J.

c. *Heat equation*

The temperature field satisfies

$$C_{\text{eff}}(J)\,\frac{\partial T}{\partial t} = \frac{\partial}{\partial X}\left(K_T(J)\,\frac{\partial T}{\partial X}\right) - \frac{\partial}{\partial X}\left(c_p^{\ell}\,T\,Q_s\right) + (-\Delta H_r)\,J\,R(c,T,J), \tag{A8}$$

where $C_{\text{eff}}(J)$ is the effective volumetric heat capacity, $K_T(J)$ is the effective thermal conductivity, c_p^{ℓ} is the solvent heat capacity per unit volume, and $-\Delta H_r > 0$ is the exothermic reaction enthalpy. The three terms on the right-hand side represent, respectively, thermal conduction, enthalpy advection by the solvent flow, and reaction heat generation.

3. Free energy and chemical potential

The Helmholtz free-energy density per unit reference volume is written as

$$\Psi(J,T) = \Psi_{\text{el}}(J) + \Psi_{\text{mix}}(\phi,T) + \frac{\kappa_J}{2}\left(\frac{\partial J}{\partial X}\right)^2, \tag{A9}$$

where $\phi = \phi_{p0}/J$, and the three contributions are the elastic energy of the network, the Flory–Huggins mixing energy, and a gradient-energy regularization.

The mixing free-energy density takes the standard Flory–Huggins form:

$$\Psi_{\text{mix}}(\phi,T) = \frac{R_g T}{v_s}\Big[(1-\phi)\ln(1-\phi) + \chi(T,\phi)\,\phi(1-\phi)\Big], \tag{A10}$$

with R_g the gas constant, v_s the solvent molar volume, and the Flory–Huggins interaction parameter

$$\chi(T,\phi) = \chi_0(T) + \chi_1\,\phi. \tag{A11}$$

The temperature-dependent part $\chi_0(T)$ encodes the lower critical solution temperature (LCST) behavior of the gel.

The solvent chemical potential is then

$$\mu_s = \mu_{\text{mix}}(J,T) + \mu_{\text{el}}(J) - \kappa_J\,\frac{\partial^2 J}{\partial X^2}, \tag{A12}$$

or equivalently a generalized swelling stress:

$$\Sigma(J,T) = \Sigma_{\text{el}}(J) - \Sigma_{\text{mix}}(J,T) - \kappa_J\,\frac{\partial^2 J}{\partial X^2}, \tag{A13}$$

with the solvent flux alternatively expressed as $Q_s = -M_s(J,T)\,\partial_X \Sigma$.

4. Reaction kinetics

The volumetric reaction rate takes the Arrhenius form modulated by a collapse-dependent accessibility factor:

$$R(c,T,J) = k_0\,c^n\,\exp\!\left(-\frac{E_a}{R_g T}\right)a(J), \tag{A14}$$

where k_0 is the pre-exponential factor, n the reaction order, and E_a the activation energy.

The activity factor $a(J)$ models the suppression of reaction due to reduced free solvent volume upon collapse:

$$a(J) = \left(1 - \frac{\phi_{p0}}{J}\right)^m = (1-\phi)^m. \tag{A15}$$

When the gel collapses ($J \to \phi_{p0}$, $\phi \to 1$), the activity vanishes, shutting off the reaction.

The effective transport coefficients similarly depend on the microstructure:

$$D_c(J) = D_0\left(1 - \frac{\phi_{p0}}{J}\right)^{m_D}, \tag{A16}$$

$$M_s(J) = M_0\left(1 - \frac{\phi_{p0}}{J}\right)^{m_M}, \tag{A17}$$

so that both solute diffusion and solvent permeation are suppressed in the collapsed state.

5. Oscillation mechanism

The coupled model gives rise to a closed feedback loop:

1. Reactant enters through the surface and is consumed, releasing heat.
2. Temperature rises, increasing χ and driving the gel toward collapse ($J \downarrow$).

3. Collapse reduces pore size, suppressing both permeation ($M_s \downarrow$) and diffusion ($D_c \downarrow$), as well as reaction accessibility ($a = (1-\phi)^{m_{\rm act}} \downarrow$).

4. The heat source is extinguished; the gel cools by conduction and surface exchange.

5. Cooling lowers χ; the gel re-swells ($J \uparrow$), reopening transport pathways.

6. Fresh reactant enters, and the cycle repeats.

Self-sustained oscillation requires sufficient time-scale separation between the fast positive feedback (reaction → heating → collapse) and the slow negative feedback (collapse → transport suppression → cooling → re-swelling).

The reaction shutoff in step 3 is spatially heterogeneous. In the outer shell ($\xi \gtrsim 0.7$), the collapse drives $a \to 0$ while the local reactant concentration is still finite ($u_{\rm surf} \approx 0.25$); shutoff is *accessibility-controlled*. In the inner core ($\xi \lesssim 0.5$), the Thiele modulus is large enough that u depletes before the core collapses; shutoff there is *reactant-depletion controlled*. Post-collapse, the surface Robin supply refills $u_{\rm surf}$ to ≈ 1 (the collapsed shell blocks consumption), "storing" reactant for the next reignition.

6. Boundary conditions

At the symmetry plane $X = 0$:

$$Q_s = 0, \qquad N_c = 0, \qquad \frac{\partial T}{\partial X} = 0. \tag{A18}$$

At the free surface $X = H_0$:

a. Solvent exchange. A Robin-type condition couples the surface chemical potential to the bath:

$$Q_s = L_\mu(\mu_s - \mu_{\rm bath})\,, \tag{A19}$$

where L_μ is a surface transfer coefficient. In the limit of rapid equilibration, $\mu_s = \mu_{\rm bath}$.

b. Reactant supply.

$$N_c = k_c\,(c - c_b), \tag{A20}$$

where c_b is the constant bath concentration.

c. Heat exchange.

$$-K_T\,\frac{\partial T}{\partial X} = h_T\,(T - T_\infty), \tag{A21}$$

where h_T is the surface heat-transfer coefficient and T_∞ the ambient temperature.

d. Mechanical boundary. The free surface is traction-free. In the J-formulation this is implicitly satisfied by the chemical-potential balance.

Appendix B: Nondimensionalization

1. Scaling

We introduce the following dimensionless variables. Length is scaled by the reference thickness:

$$X = H_0\,\tilde{x}, \qquad \tilde{x} \in [0,1]. \tag{B1}$$

The swelling ratio J is already dimensionless. Reactant concentration is normalized by the bath value:

$$c = c_b\,u. \tag{B2}$$

Temperature is measured relative to ambient, scaled by the adiabatic reaction temperature rise:

$$T = T_\infty + \Delta T_*\,\theta, \qquad \Delta T_* = \frac{(-\Delta H_r)\,c_b}{C_*}, \tag{B3}$$

where C_* is a representative volumetric heat capacity.

The chemical potential is scaled by the thermal mixing energy:

$$\mu_s = \mu_*\,m, \qquad \mu_* = \frac{R_g T_\infty}{v_s}. \tag{B4}$$

Time is scaled by the swelling/permeation time:

$$\tau_s = \frac{H_0^2}{M_0\,\mu_*}, \tag{B5}$$

where M_0 is a reference mobility. The solvent and reactant flux scales follow:

$$Q_s = \frac{H_0}{\tau_s}\,q, \qquad N_c = \frac{c_b\,H_0}{\tau_s}\,n. \tag{B6}$$

Material functions are factored as reference value times dimensionless function:

$$M_s = M_0\,\mathcal{M}(J,\theta), \qquad D_c = D_0\,\mathcal{D}(J,\theta), \tag{B7}$$

$$C_{\rm eff} = C_*\,\mathcal{C}(J), \qquad K_T = K_*\,\mathcal{K}(J). \tag{B8}$$

2. Dimensionless reaction rate

Defining the reference reaction rate

$$R_0 = k_0\,c_b^n\,\exp\!\left(-\frac{E_a}{R_g T_\infty}\right), \tag{B9}$$

we write $R = R_0\,\widehat{R}(u,\theta,J)$ with

$$\widehat{R}(u,\theta,J) = u^n\,a(J)\,\exp\!\left[\frac{\Gamma_A\,\theta}{1+\varepsilon_T\,\theta}\right], \tag{B10}$$

where

$$\varepsilon_T = \frac{\Delta T_*}{T_\infty}, \qquad \Gamma_A = \frac{E_a\,\Delta T_*}{R_g\,T_\infty^2}. \tag{B11}$$

When $\varepsilon_T \ll 1$, this simplifies to $\widehat{R} \approx u^n\,a(J)\,e^{\Gamma_A\theta}$.

3. Dimensionless chemical potential

The dimensionless chemical potential is

$$m = m_{\rm mix}(J,\theta) + m_{\rm el}(J) - \ell^2\,\frac{\partial^2 J}{\partial \tilde{x}^2}, \tag{B12}$$

with the interface parameter

$$\ell^2 = \frac{\kappa_J}{\mu_*\,H_0^2}. \tag{B13}$$

The mixing part retains the Flory–Huggins form:

$$m_{\rm mix} = \ln(1-\phi) + \phi + \chi(\theta,\phi)\,\phi^2, \qquad \phi = \frac{\phi_{p0}}{J}, \tag{B14}$$

with the dimensionless interaction parameter

$$\chi(\theta,\phi) = \chi_\infty + S_\chi\,\theta + \chi_1\,\phi, \tag{B15}$$

where $\chi_\infty = \chi_0(T_\infty)$ and the thermo-swelling sensitivity is

$$S_\chi = \Delta T_*\,\left.\frac{d\chi_0}{dT}\right|_{T_\infty}. \tag{B16}$$

The elastic contribution is

$$m_{\rm el} = \Omega_e\,\widehat{m}_{\rm el}(J), \qquad \Omega_e = \frac{G\,v_s}{R_g\,T_\infty}, \tag{B17}$$

measuring the network stiffness relative to the thermal mixing energy.

4. Dimensionless governing equations

Dropping tildes for clarity, the dimensionless system reads:

a. Swelling equation.

$$\boxed{\frac{\partial J}{\partial t} = \frac{\partial}{\partial x}\left(\mathcal{M}(J,\theta)\,\frac{\partial m}{\partial x}\right)} \tag{B18}$$

b. Chemical potential.

$$\boxed{m = m_{\rm mix}(J,\theta) + m_{\rm el}(J) - \ell^2\,\frac{\partial^2 J}{\partial x^2}} \tag{B19}$$

c. Reactant equation.

$$\boxed{\frac{\partial (Ju)}{\partial t} + \frac{\partial}{\partial x}\left(u\,q - \delta\,\mathcal{D}(J,\theta)\,\frac{\partial u}{\partial x}\right) = -{\rm Da}\,J\,\widehat{R}(u,\theta,J)} \tag{B20}$$

where $q = -\mathcal{M}(J,\theta)\,\partial_x m$ and

$$\delta = \frac{D_0}{M_0\,\mu_*} \tag{B21}$$

is the ratio of solute diffusion speed to solvent permeation speed.

The Damköhler number, measuring reaction rate relative to swelling rate, is

$$\mathrm{Da} = \tau_s\,k_0\,c_b^{n-1}\,\exp\!\left(-\frac{E_a}{R_g T_\infty}\right). \tag{B22}$$

d. Heat equation.

$$\boxed{\mathcal{C}(J)\,\frac{\partial \theta}{\partial t} = \alpha\,\frac{\partial}{\partial x}\left(\mathcal{K}(J)\,\frac{\partial \theta}{\partial x}\right) - {\rm Pe}_T\,\frac{\partial}{\partial x}[(\Theta_\infty + \theta)\,q] + {\rm Da}\,J\,\widehat{R}(u,\theta,J} \tag{B23}$$

with

$$\alpha = \frac{K_*\,\tau_s}{C_*\,H_0^2} = \frac{\tau_s}{\tau_T}, \qquad \tau_T = \frac{C_*\,H_0^2}{K_*}, \tag{B24}$$

$$\mathrm{Pe}_T = \frac{c_p^\ell\,M_0\,\mu_*}{K_*}, \tag{B25}$$

$$\Theta_\infty = \frac{T_\infty}{\Delta T_*}. \tag{B26}$$

When the enthalpy-advection Péclet number is small ($\mathrm{Pe}_T \ll 1$), the convective term may be dropped.

e. Constraint.

$$\phi = \frac{\phi_{p0}}{J}. \tag{B27}$$

5. Dimensionless boundary conditions

At $x = 0$ (symmetry):

$$q = 0, \qquad u\,q - \delta\,\mathcal{D}\,u_x = 0, \qquad \theta_x = 0. \tag{B28}$$

At $x = 1$ (free surface):

$$q = \mathrm{Bi}_\mu\,(m - m_b), \tag{B29}$$

$$u\,q - \delta\,\mathcal{D}\,u_x = \mathrm{Bi}_c\,(u-1), \tag{B30}$$

$$-\mathcal{K}(J)\,\theta_x = \mathrm{Bi}_T\,\theta, \tag{B31}$$

where the three Biot numbers are

$$\mathrm{Bi}_\mu = \frac{L_\mu\,H_0}{M_0}, \qquad \mathrm{Bi}_c = \frac{k_c\,H_0}{M_0\,\mu_*}, \qquad \mathrm{Bi}_T = \frac{h_T\,H_0}{K_*}. \tag{B32}$$

6. Controlling parameter groups

The oscillatory dynamics are governed by the following dimensionless groups:

- Da — Damköhler number: strength of reaction heat source relative to swelling.
- $\alpha = \tau_s/\tau_T$ — thermal diffusion relative to swelling.
- S_χ — thermo-swelling sensitivity: how strongly temperature shifts the interaction parameter.
- Γ_A — Arrhenius feedback strength.
- δ — solute diffusion to permeation ratio.
- ℓ — interface regularization length.

- Bi_c, Bi_T — surface exchange rates for reactant and heat.

Together with the working-point parameters ϕ_{p0} and χ_∞, these form the minimal set controlling the onset of self-oscillation. In physical terms, oscillation requires that the fast positive feedback (reaction → heating → collapse) and the slow negative feedback (collapse → transport suppression → cooling → re-swelling) operate on well-separated time scales.

Appendix C: Numerical Method

We describe a cell-centered finite-volume discretization with implicit–explicit (IMEX) time stepping for the dimensionless system derived in Appendix B. The approach casts the fourth-order swelling equation in mixed form and employs upwind advection for the reactant transport.

1. Mixed formulation

To avoid discretizing fourth-order derivatives directly, we introduce the auxiliary variables m, q, n, and h and write the system as

$$J_t + q_x = 0, \qquad q = -\mathcal{M}(J,\theta)\, m_x, \tag{C1}$$

$$m = f(J,\theta) - \ell^2\, J_{xx}, \tag{C2}$$

$$(Ju)_t + n_x = -\mathrm{Da}\, J\, \widehat{R}, \qquad n = u\, q - \delta\, \mathcal{D}(J,\theta)\, u_x, \tag{C3}$$

$$\mathcal{C}(J)\, \theta_t + h_x = \mathrm{Da}\, J\, \widehat{R}, \qquad h = -\alpha\, \mathcal{K}(J)\, \theta_x, \tag{C4}$$

where $f(J,\theta) = m_{\mathrm{mix}}(J,\theta) + m_{\mathrm{el}}(J)$. This mixed form requires only C^0 regularity (no second-derivative continuity) and naturally accommodates Robin boundary conditions.

2. Spatial discretization

We use a uniform mesh of N cells:

$$\Delta x = \frac{1}{N}, \qquad x_i = \left(i - \tfrac{1}{2}\right)\Delta x, \quad i = 1, \ldots, N. \tag{C5}$$

Face positions are $x_{i+1/2} = i\,\Delta x$ for $i = 0, \ldots, N$. Cell-centered unknowns are J_i, m_i, u_i, θ_i; face fluxes are $q_{i+1/2}$, $n_{i+1/2}$, $h_{i+1/2}$.

a. Swelling equation

The conservative update reads

$$\frac{J_i^{n+1} - J_i^n}{\Delta t} + \frac{q_{i+1/2}^{n+1} - q_{i-1/2}^{n+1}}{\Delta x} = 0, \tag{C6}$$

with the face flux

$$q_{i+1/2}^{n+1} = -\mathcal{M}_{i+1/2}^{*}\, \frac{m_{i+1}^{n+1} - m_i^{n+1}}{\Delta x}. \tag{C7}$$

The face mobility $\mathcal{M}_{i+1/2}^{*}$ is evaluated using the harmonic mean

$$\mathcal{M}_{i+1/2}^{*} = \frac{2\,\mathcal{M}_i^{*}\,\mathcal{M}_{i+1}^{*}}{\mathcal{M}_i^{*} + \mathcal{M}_{i+1}^{*}}, \tag{C8}$$

which is more robust than arithmetic averaging for degenerate mobilities.

b. Chemical potential

The discrete chemical potential is

$$m_i^{n+1} = f(J_i^{n+1}, \theta_i^{*}) - \ell^2\, \frac{J_{i+1}^{n+1} - 2J_i^{n+1} + J_{i-1}^{n+1}}{\Delta x^2}, \tag{C9}$$

where the natural boundary condition $\partial_x J = 0$ is imposed via ghost cells:

$$J_0^{n+1} = J_1^{n+1}, \qquad J_{N+1}^{n+1} = J_N^{n+1}. \tag{C10}$$

c. Reactant equation

To preserve conservation, we update the product $W_i = J_i\, u_i$:

$$\frac{W_i^{n+1} - W_i^n}{\Delta t} + \frac{n_{i+1/2}^{n+1} - n_{i-1/2}^{n+1}}{\Delta x} = -\mathrm{Da}\, J_i^{\sharp}\, \widehat{R}(u_i^{\sharp}, \theta_i^{\sharp}, J_i^{\sharp}), \tag{C11}$$

followed by reconstruction $u_i^{n+1} = W_i^{n+1}/J_i^{n+1}$.

The reactant face flux combines upwind advection and central diffusion:

$$n_{i+1/2}^{n+1} = q_{i+1/2}^{n+1}\, u_{i+1/2}^{\mathrm{up}} - \delta\, \mathcal{D}_{i+1/2}^{*}\, \frac{u_{i+1}^{n+1} - u_i^{n+1}}{\Delta x}, \tag{C12}$$

where the upwind value is

$$u_{i+1/2}^{\mathrm{up}} = \begin{cases} u_i^{n+1}, & q_{i+1/2}^{n+1} \geq 0, \\ u_{i+1}^{n+1}, & q_{i+1/2}^{n+1} < 0. \end{cases} \tag{C13}$$

Upwinding is essential to suppress spurious oscillations when advection dominates over diffusion.

d. Heat equation

The thermal update is

$$\mathcal{C}_i^{*}\, \frac{\theta_i^{n+1} - \theta_i^n}{\Delta t} + \frac{h_{i+1/2}^{n+1} - h_{i-1/2}^{n+1}}{\Delta x} = \mathrm{Da}\, J_i^{\sharp}\, \widehat{R}(u_i^{\sharp}, \theta_i^{\sharp}, J_i^{\sharp}), \tag{C14}$$

with the heat flux (neglecting enthalpy advection)

$$h_{i+1/2}^{n+1} = -\alpha\, \mathcal{K}_{i+1/2}^{*}\, \frac{\theta_{i+1}^{n+1} - \theta_i^{n+1}}{\Delta x}. \tag{C15}$$

3. Boundary conditions

At the left boundary ($x = 0$):

$$q_{1/2}^{n+1} = 0, \qquad n_{1/2}^{n+1} = 0, \qquad h_{1/2}^{n+1} = 0. \tag{C16}$$

At the right boundary ($x = 1$), the Robin conditions become face fluxes:

$$q_{N+1/2}^{n+1} = \mathrm{Bi}_\mu\,(m_N^{n+1} - m_b), \tag{C17}$$

$$n_{N+1/2}^{n+1} = \mathrm{Bi}_c\,(u_N^{n+1} - 1), \tag{C18}$$

$$h_{N+1/2}^{n+1} = \mathrm{Bi}_T\,\theta_N^{n+1}. \tag{C19}$$

4. Time integration: IMEX splitting

Each time step proceeds in three sequential stages:

a. Stage 1: Swelling (J, m, q). Given (J^n, u^n, θ^n), solve the coupled nonlinear system (C6)–(C9) for $(J^{n+1}, m^{n+1}, q^{n+1})$ using Newton iteration. The mobility and thermodynamic function f are evaluated at the current iterate $(J^{(k)}, \theta^{(k)})$.

b. Stage 2: Reactant (W, u). With J^{n+1} and q^{n+1} known, solve the linear (or mildly nonlinear) system (C11)–(C12) for W^{n+1}, then recover $u^{n+1} = W^{n+1}/J^{n+1}$. Diffusion is treated implicitly; the reaction source may be linearized about the current temperature.

c. Stage 3: Temperature (θ). With J^{n+1} and u^{n+1} known, solve the implicit heat equation (C14)–(C15) for θ^{n+1}. If the Arrhenius source term is stiff, a Newton linearization of the source with respect to θ is applied.

5. Fully coupled Newton formulation

For problems requiring tight coupling, the unknowns are assembled into a single vector

$$\boldsymbol{Y} = (J_1, \ldots, J_N,\ m_1, \ldots, m_N,\ W_1, \ldots, W_N,\ \theta_1, \ldots, \theta_N)^T \tag{C20}$$

and the residual vector $\boldsymbol{R}(\boldsymbol{Y})$ collects all discrete equations (C6)–(C15) with boundary conditions (C16)–(C19). The system is solved by Newton's method:

$$\boldsymbol{J}_{\mathrm{ac}}\,\delta\boldsymbol{Y} = -\boldsymbol{R}(\boldsymbol{Y}^{(k)}), \qquad \boldsymbol{Y}^{(k+1)} = \boldsymbol{Y}^{(k)} + \delta\boldsymbol{Y}, \tag{C21}$$

where $\boldsymbol{J}_{\mathrm{ac}}$ is the Jacobian matrix. Because all couplings are local in one dimension, the Jacobian is banded (bandwidth ~ 4) and can be factored efficiently.

6. Implementation notes

1. The conserved variable $W = Ju$ must be used for the reactant update, not a non-conservative discretization of u alone. This ensures that boundary supply and internal reaction remain in balance even during rapid swelling or collapse.
2. Upwind discretization of the advective flux $u\,q$ is mandatory, not optional. Central differencing produces non-physical oscillations whenever $|q|\,\Delta x/(\delta\,\mathcal{D}) \gg 1$.
3. The fourth-order swelling–chemical-potential system must be split into two second-order equations (mixed form). Direct discretization of J_{xxxx} leads to wider stencils and difficulties with natural boundary conditions.